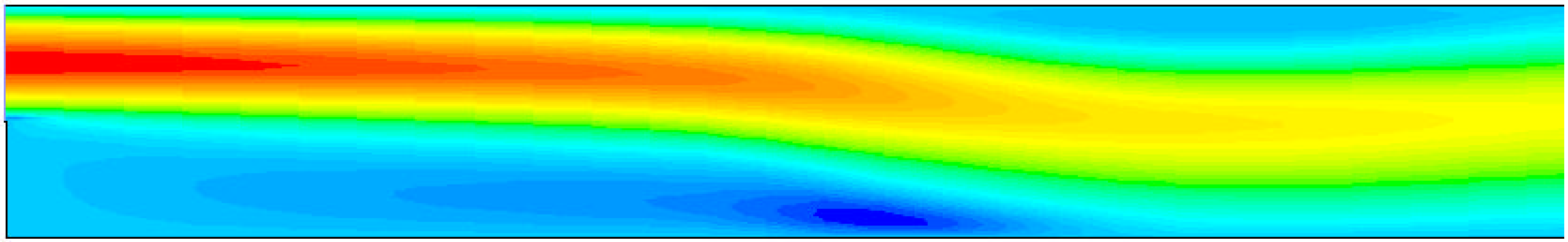
CFD Benchmark Testing of Selected Laminar Flow Problems



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Outline

- Background
- Motivation and Goals
- Problem description, Results and Discussion
  - Developing flow in a channel
  - Thermally developing slug flow in a channel
  - Flow over a backward-facing step
  - Mixed convection in a heated vertical duct with a step
  - Natural convection in a square cavity
- Summary
- Future Work

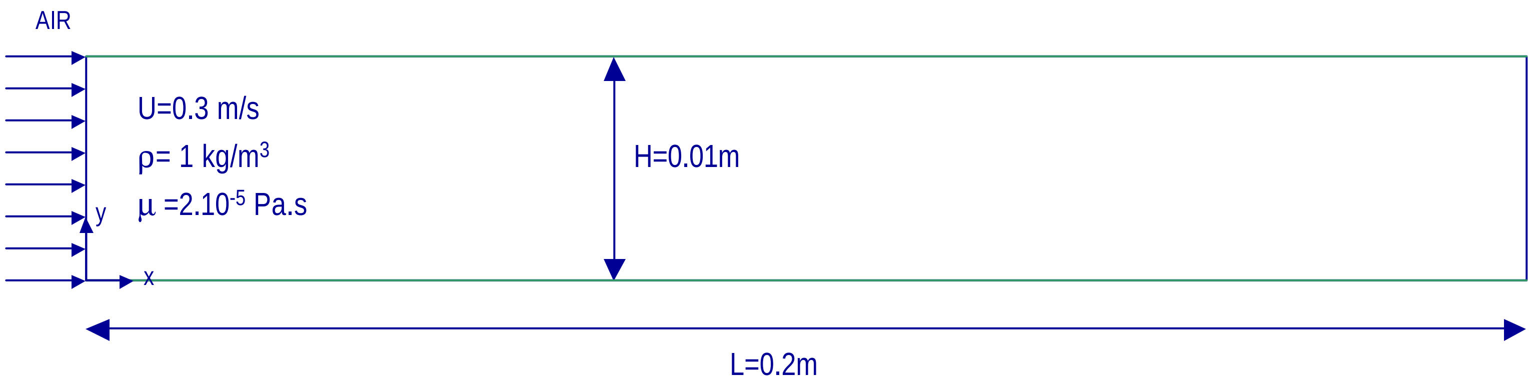
Background

- CFD extensively and successfully used for predicting flow and associated heat transfer
- A CFD code should incorporate
  - A certain mathematical formulation
  - Turbulence model
  - Numerical scheme
  - Solution algorithm
- Results are obtained by specifying
  - Required boundary conditions
  - Material properties
  - Grid sizes

Motivation and Goals

- Benchmark problem:
  - For the developer of a CFD code → a good opportunity to check and evaluate the code
  - For the user of a CFD code → direct way to judge satisfactoriness of the code
  - Solution algorithm
- Both the developers and users can employ the benchmark testing to develop guidelines for the degree of grid refinement for a given accuracy.
- Selected benchmark problems provide a good test for the CFD code to see (FLUENT)
  - treatment of pressure-velocity and velocity-temperature couplings
  - entrance region effects
  - Bouyancy effects

Developing flow in a 2D channel



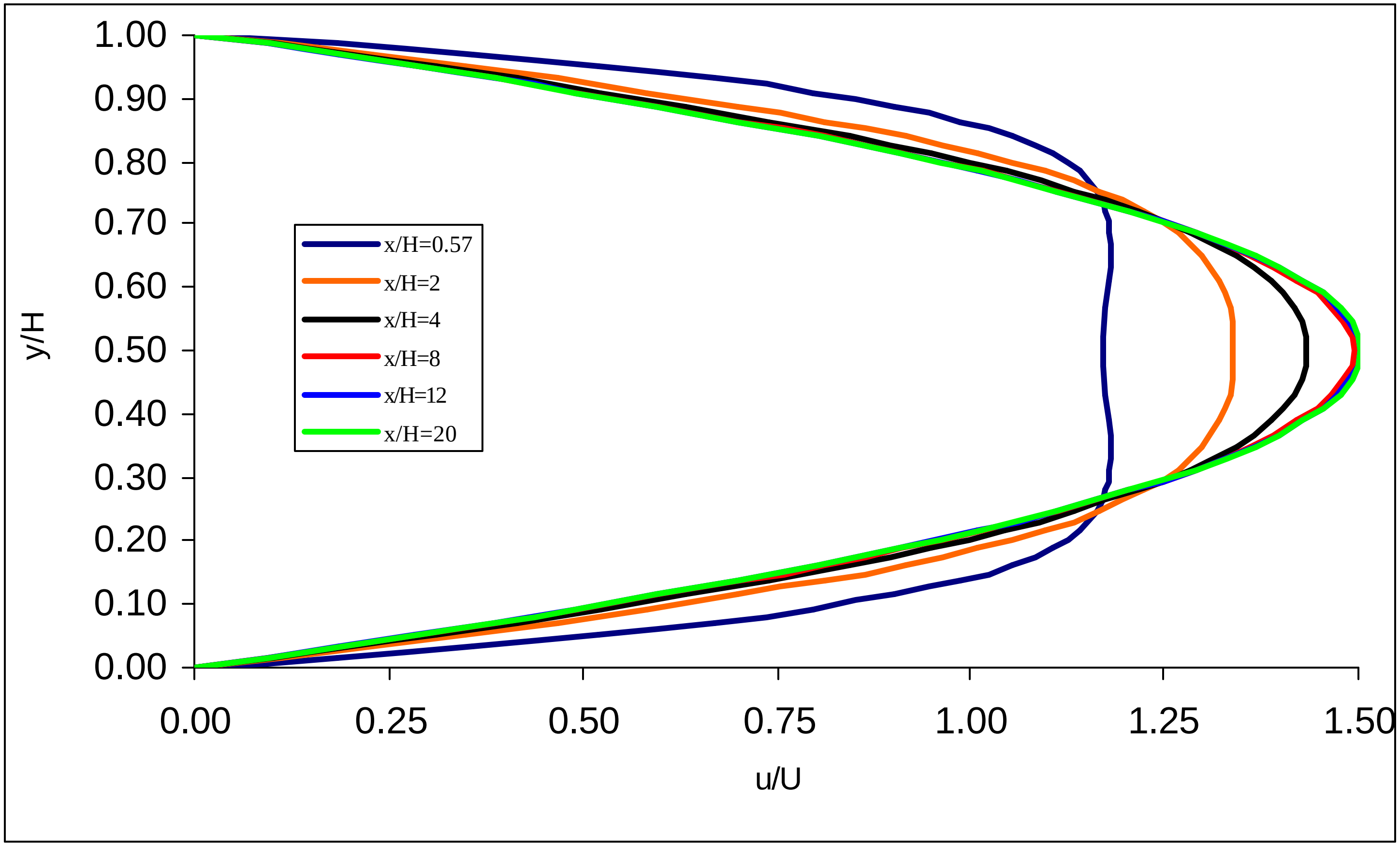
- Laminar flow, Re_H=150
- Incompressible flow
- Steady flow
- Constant properties

Developing flow in a 2D channel

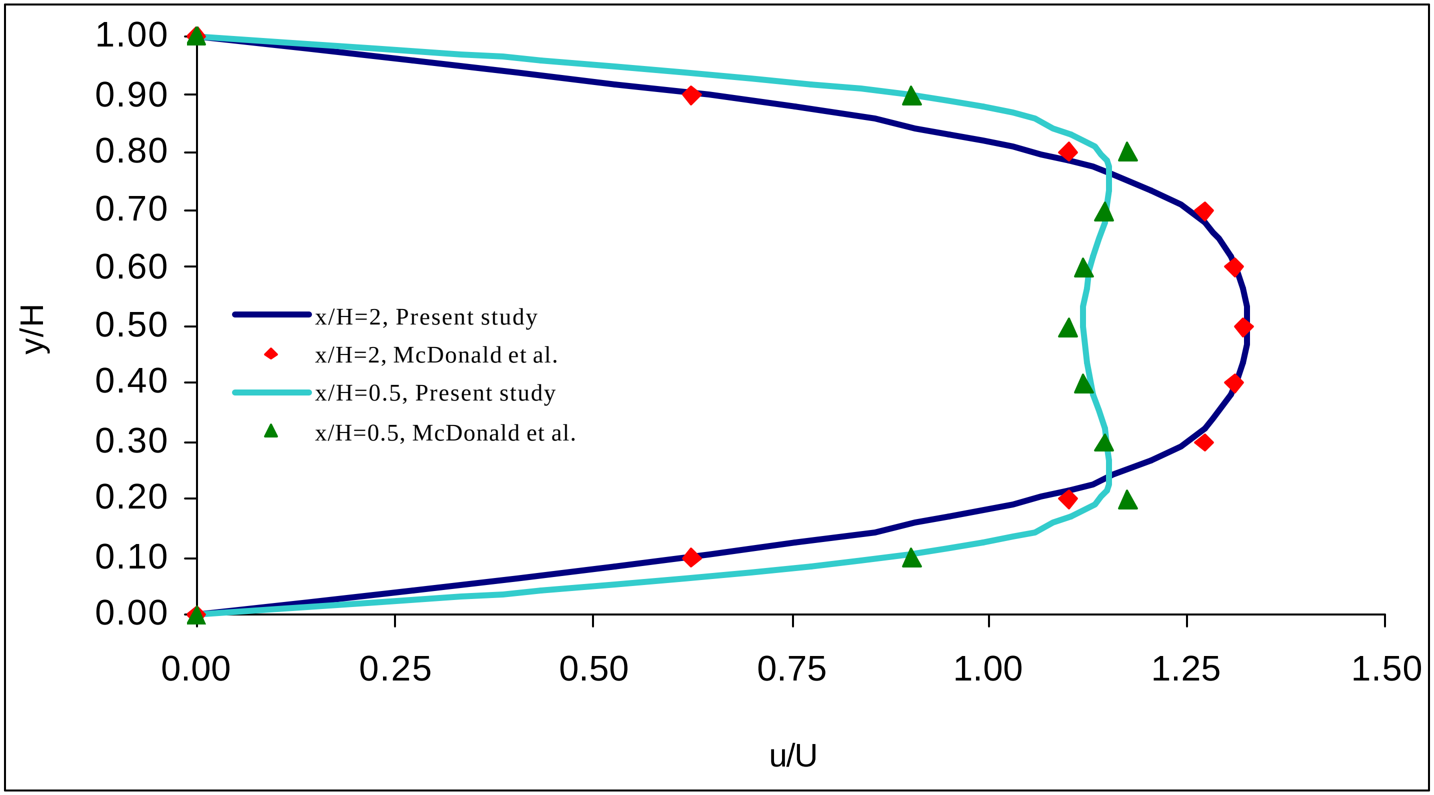
Grid size	Meshing	# iterations	Grid size	Meshing	# iterations
50*20	uniform	34	80*50	uniform	102
50*30	uniform	53	80*50	non-uniform	83
50*30	non-uniform	45	80*35	non-uniform	57
70*35	uniform	66	200*20	non-uniform	45
70*35	non-uniform	56			

Residual	conv. cr.	Residual	conv. cr.
x-momentum	1.00E-04	continuity	1.00E-04
y-momentum	1.00E-04	energy	1.00E-06

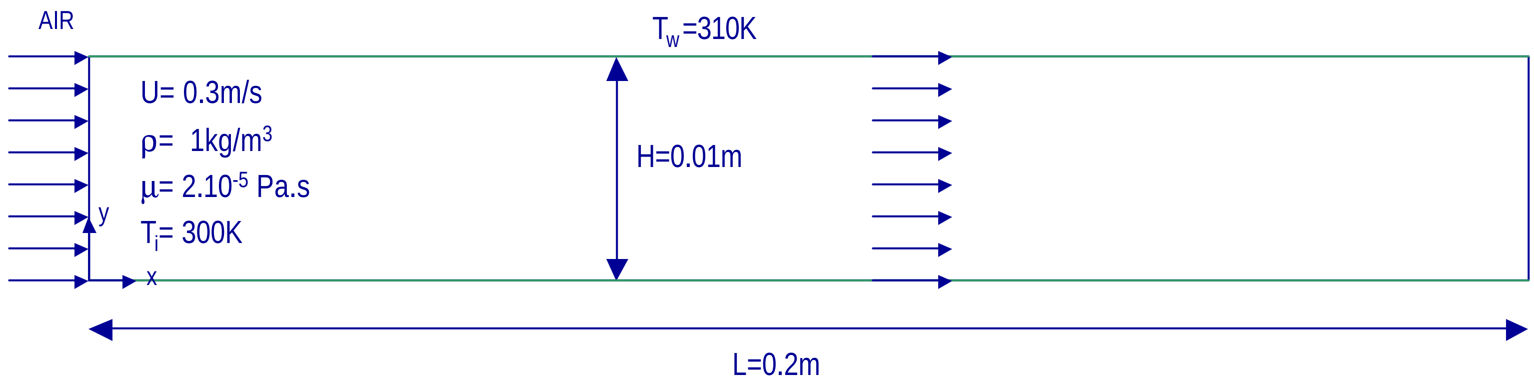
Developing flow in a 2D channel



Developing flow in a 2D channel

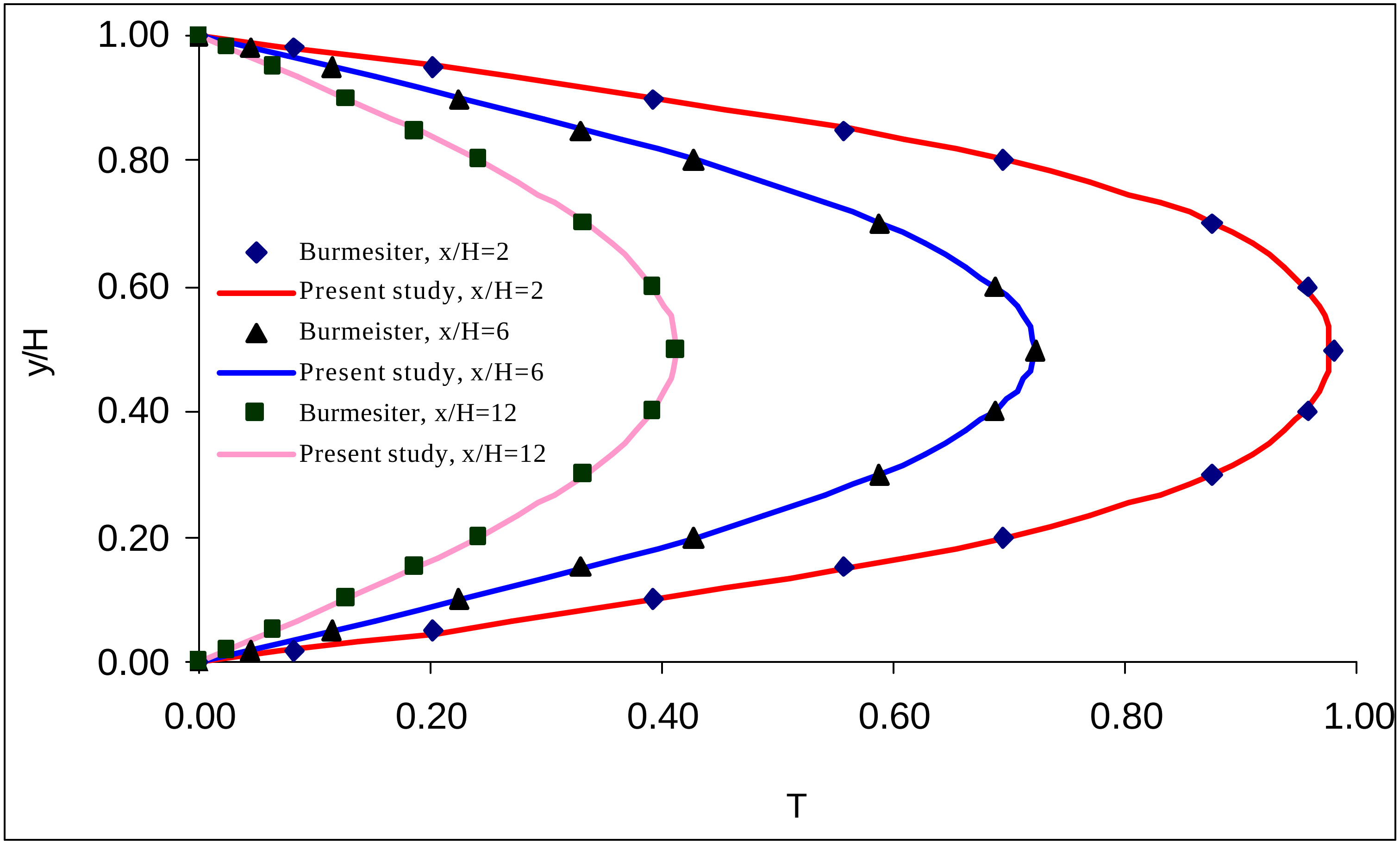


Thermally developing slug-flow in a 2D channel

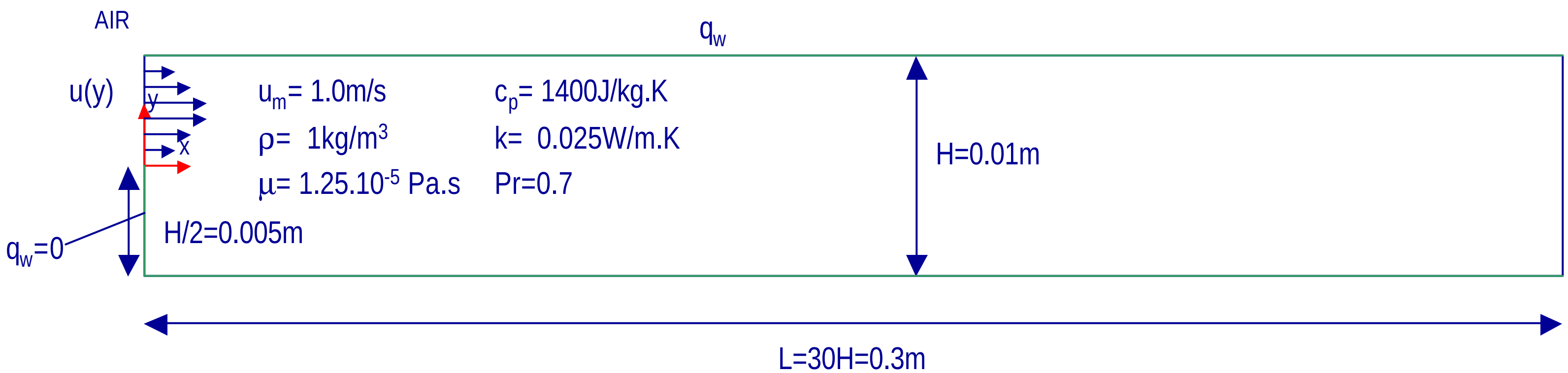


- Slug flow
- Laminar flow, Re_H=150
- Incompressible flow
- Steady flow
- Constant properties

Thermally developing slug-flow in a 2D channel



Flow over a backward facing step



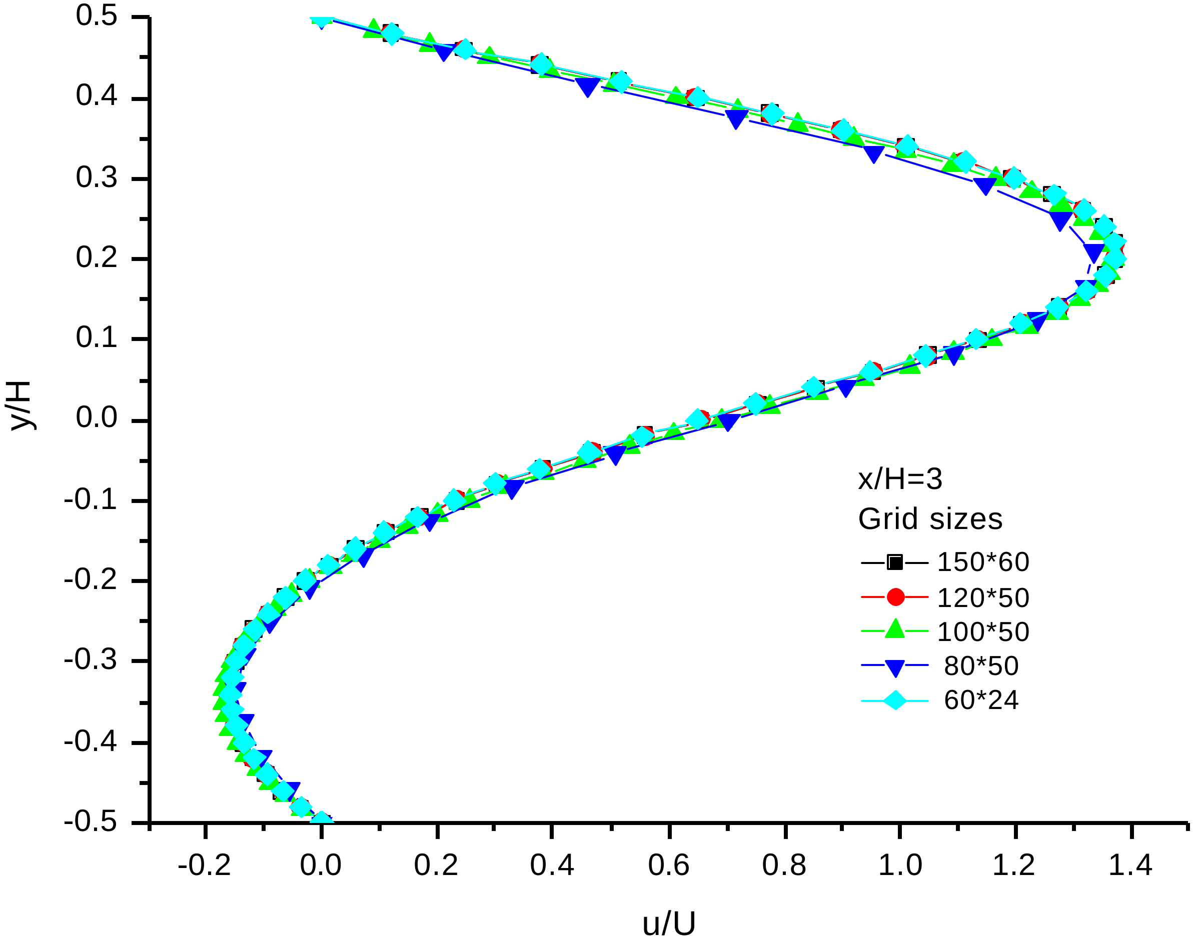
- Laminar flow, Re_H=800
- Incompressible flow
- Steady flow
- Constant properties

- $T_{cl,i}$=300K, $T_{w,i}$=310K

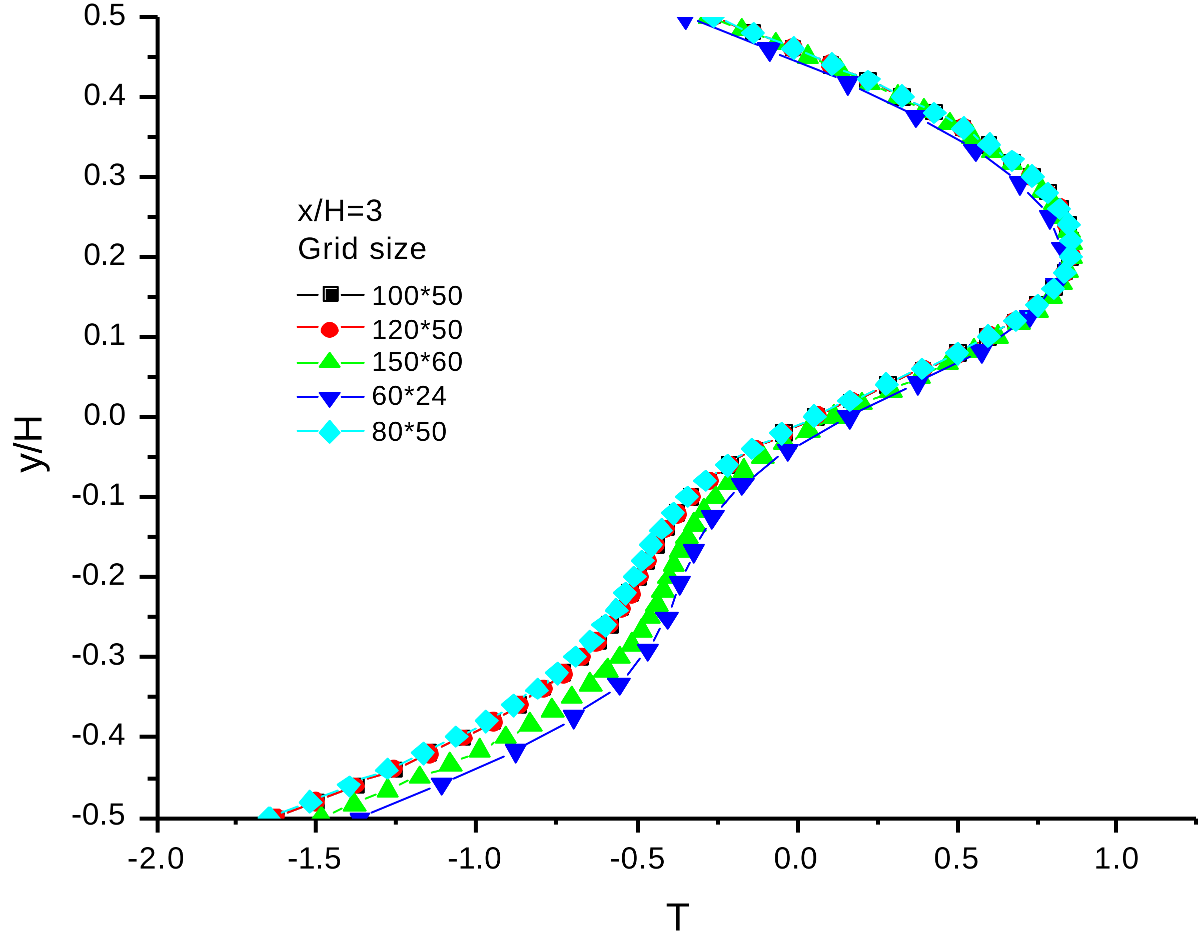
$$q''_w = \frac{32}{5}\frac{k}{H}\left(T_{w,i} - T_{cl,i}\right)$$

$$\frac{T(y) - T_{w,i}}{T_{cl,i} - T_{w,i}} = \left[1 - \left(\frac{4y}{H} - 1\right)^2\right]\left[1 - \frac{1}{5}\left(\frac{4y}{H} - 1\right)^2\right]$$

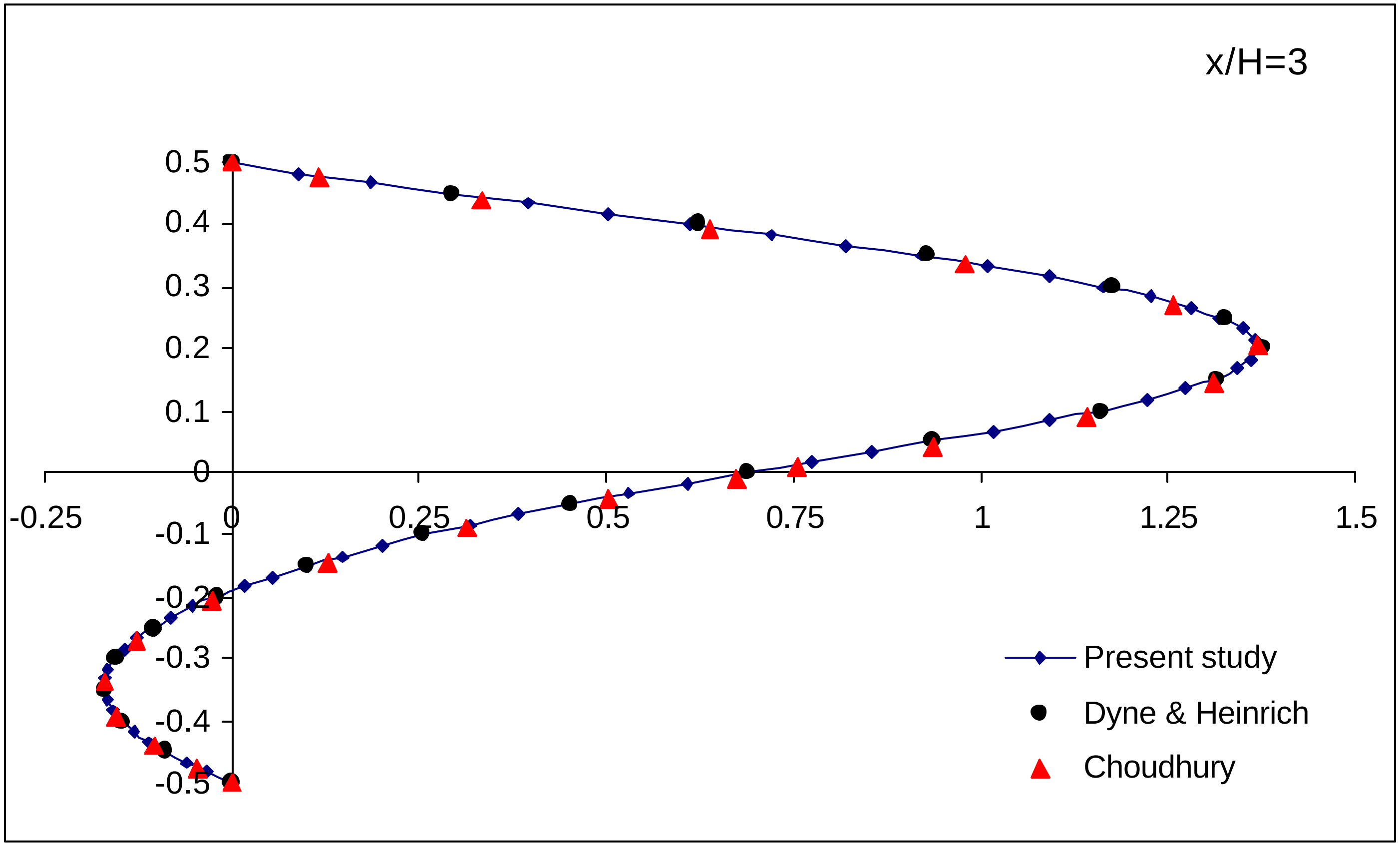
Flow over a backward facing step



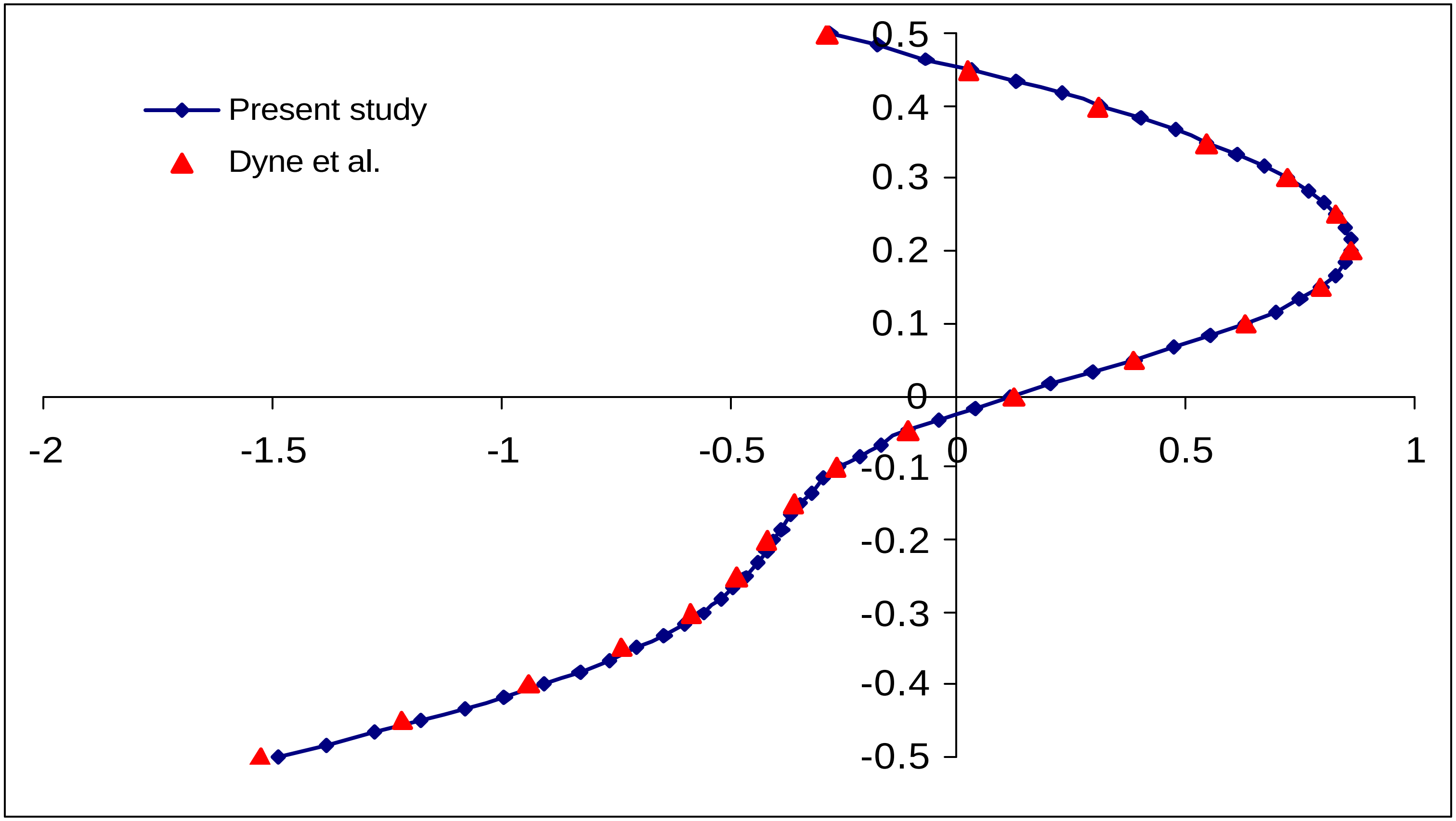
Flow over a backward facing step



Flow over a backward facing step



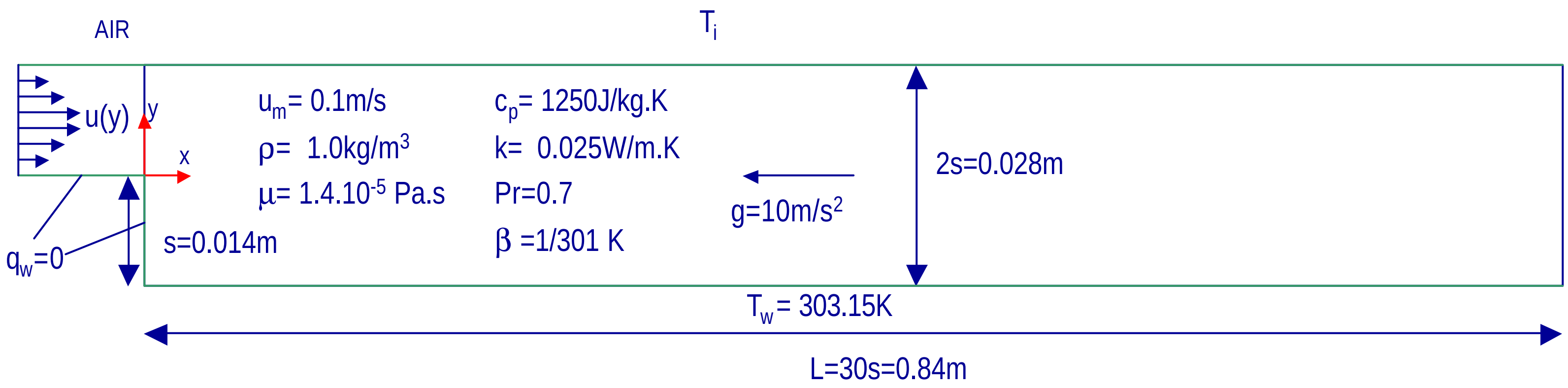
Flow over a backward facing step



Flow over a backward facing step

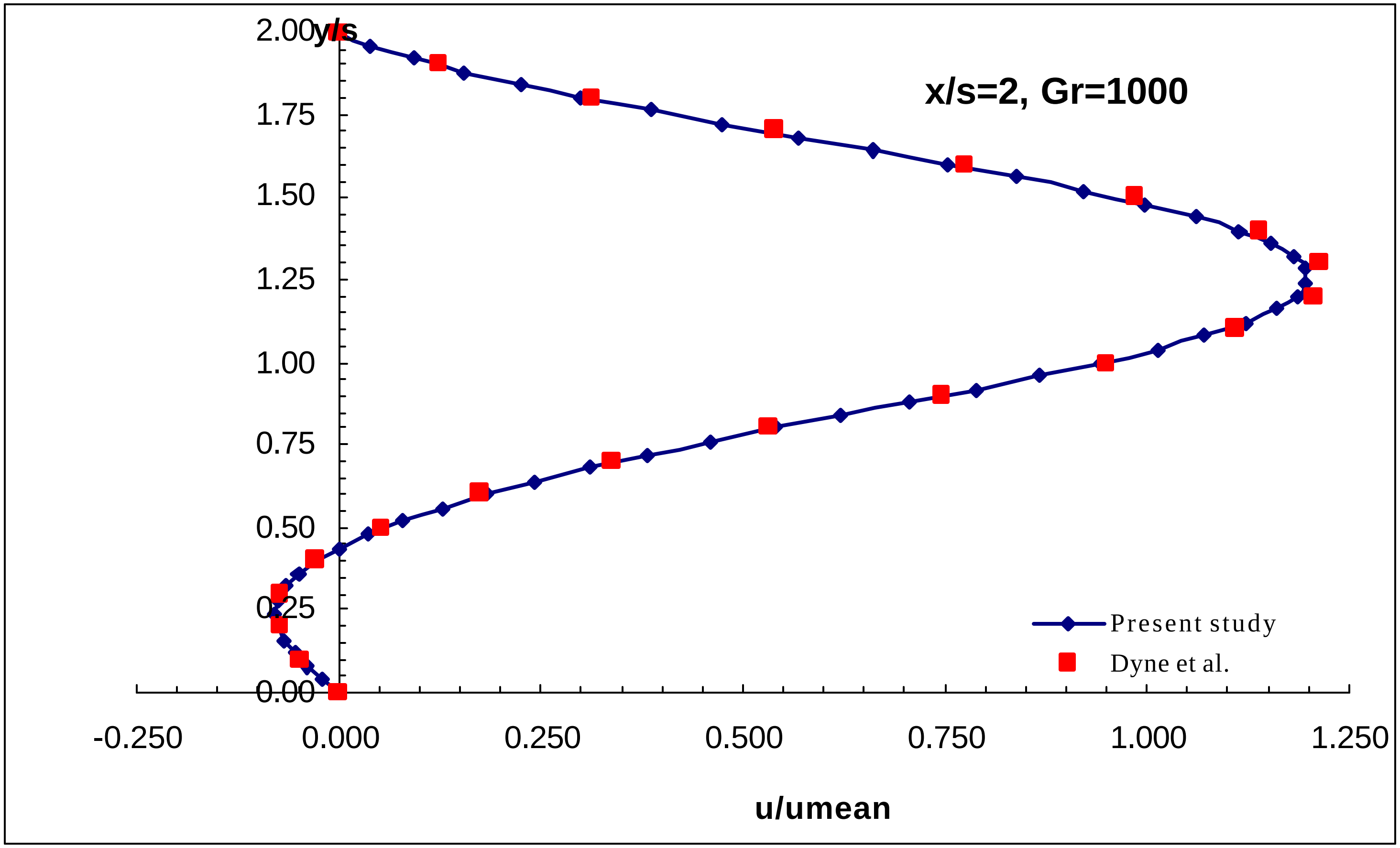
	x/H			
	Dogruoz	Dyne	Choudhury	Gartling
Lower wall reattachment point	6.0	5.925	5.80	6.10
Upper wall seperation point	4.8	4.925	4.65	4.85
Upper wall reattachment point	10.4	10.225	10.35	10.48

Mixed convection flow over a vertical backward facing step

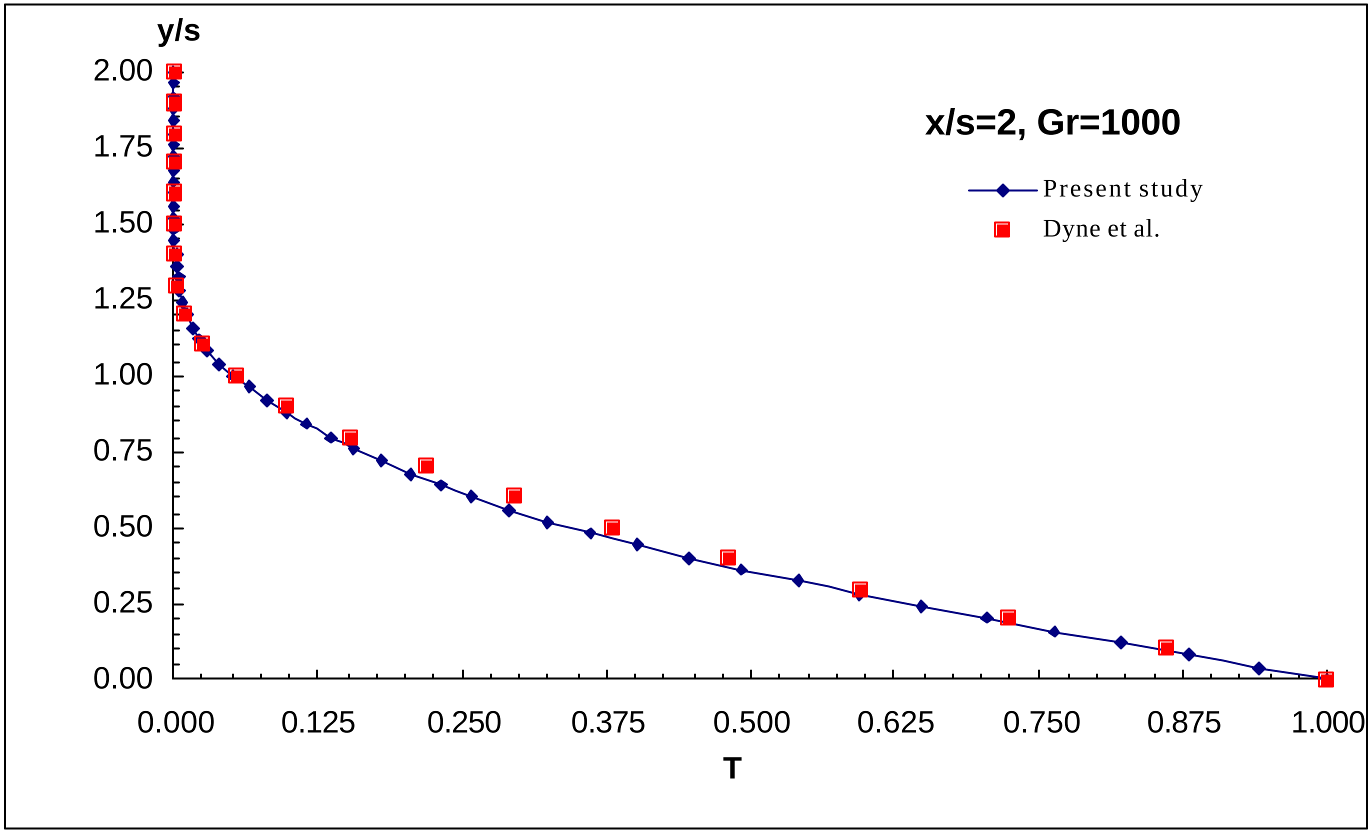


- Laminar flow, Re_s=100
- Incompressible flow
- Steady flow
- Constant properties
- T_w=303.15K, T_i=301K
- Gr_s=1000, Pe_s=70

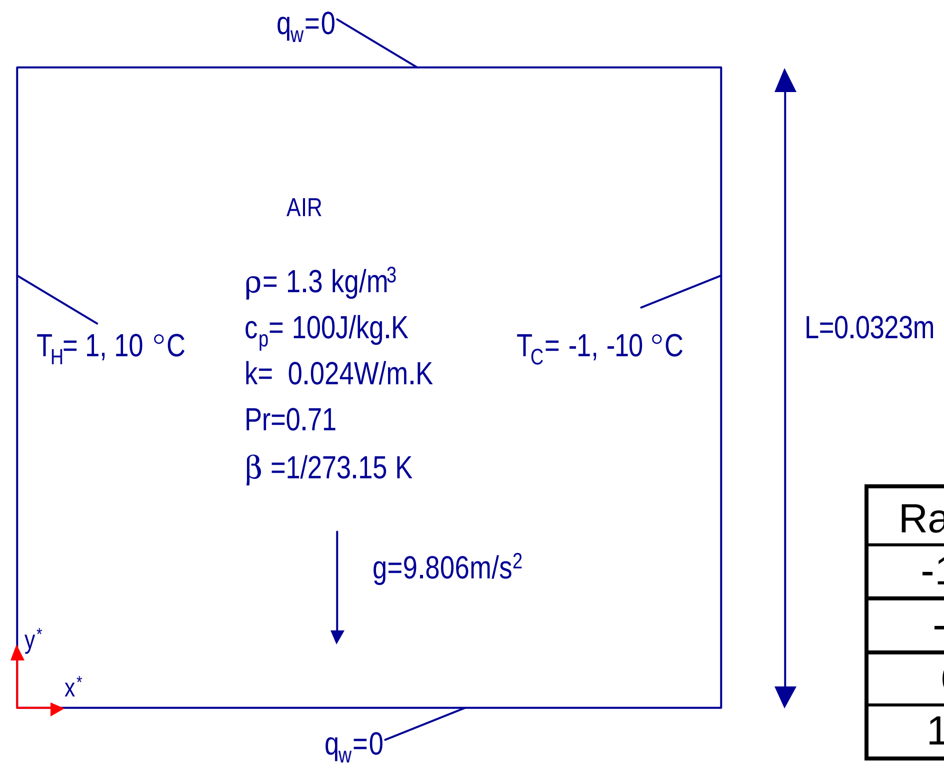
Mixed convection flow over a vertical backward facing step



Mixed convection flow over a vertical backward facing step



Buoyancy-driven flow in a square enclosure with variable viscosity



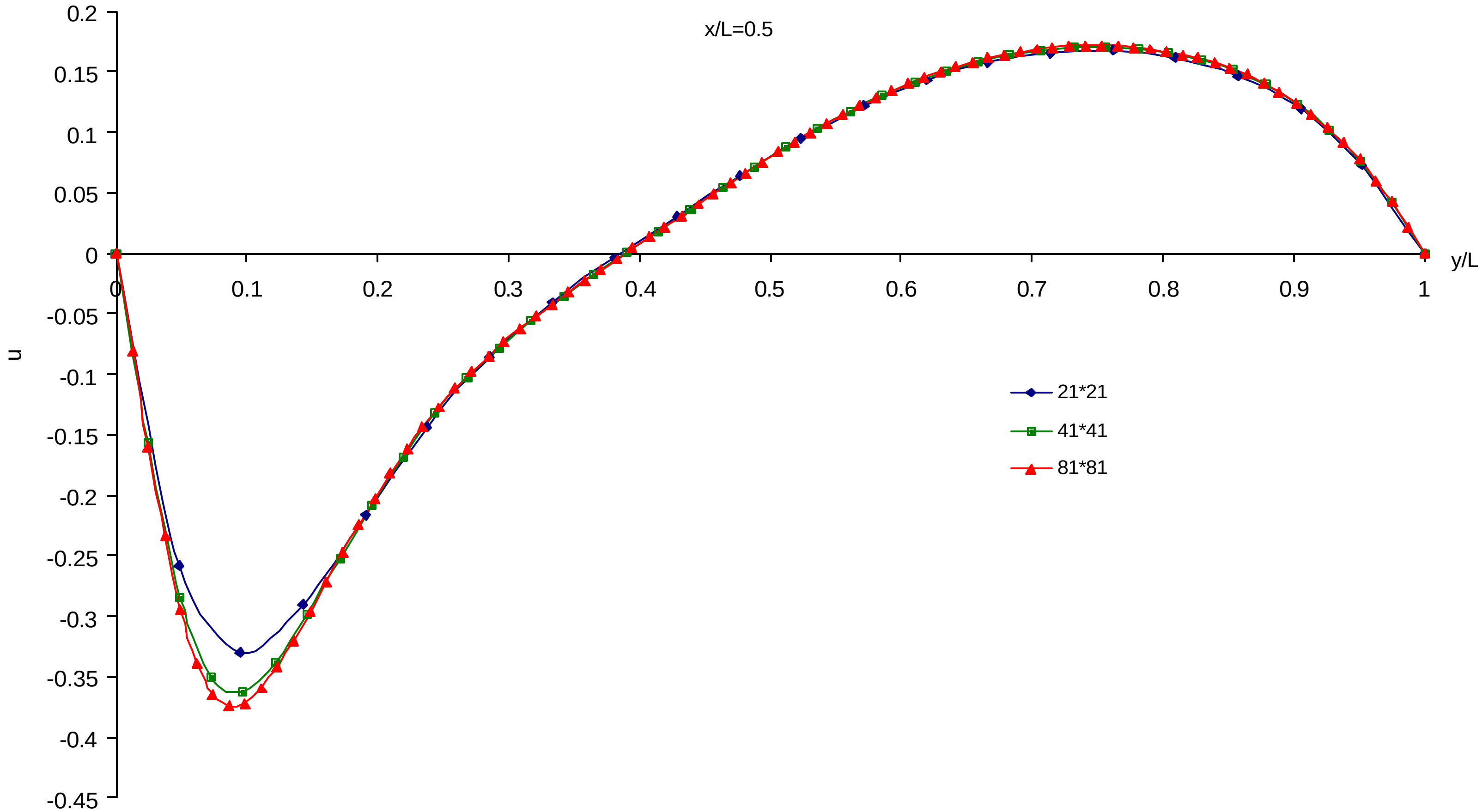
$$x = \frac{x^*}{L} \qquad T = \frac{T^* - T_0^*}{\Delta T^*}$$

$$u = \frac{u^*}{\sqrt{Ra_0\, Pr_0}\,\dfrac{a_0}{L}}$$

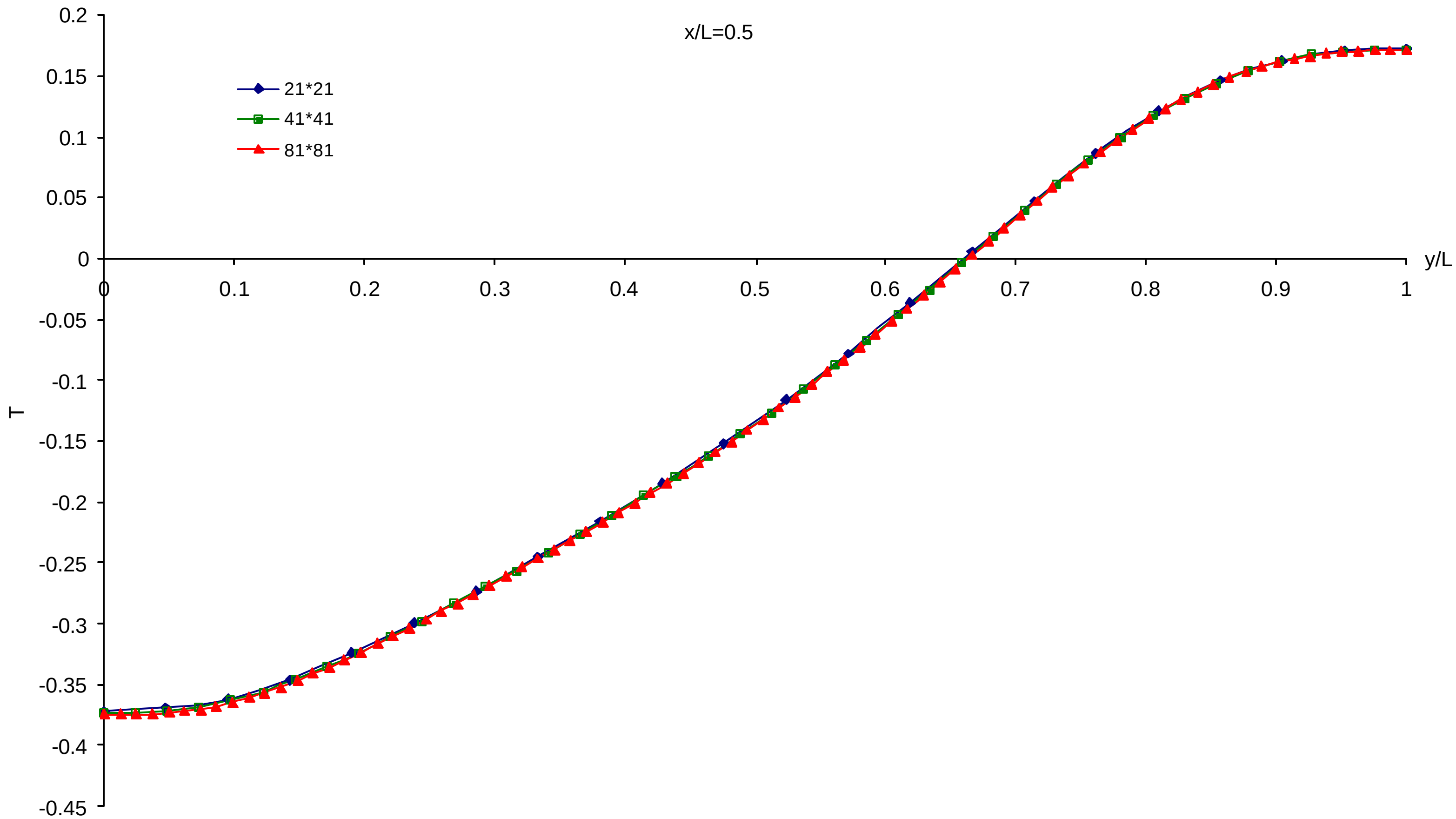
- $Ra_0 = 10^4, 10^5$

Ra=10^5	Ra=10^4	Viscosity (Pa.s)
-10.0	-1.0	3.40E-07
-6.0	-0.6	3.40E-06
6.0	0.6	3.06E-05
10.0	1.0	3.40E-04

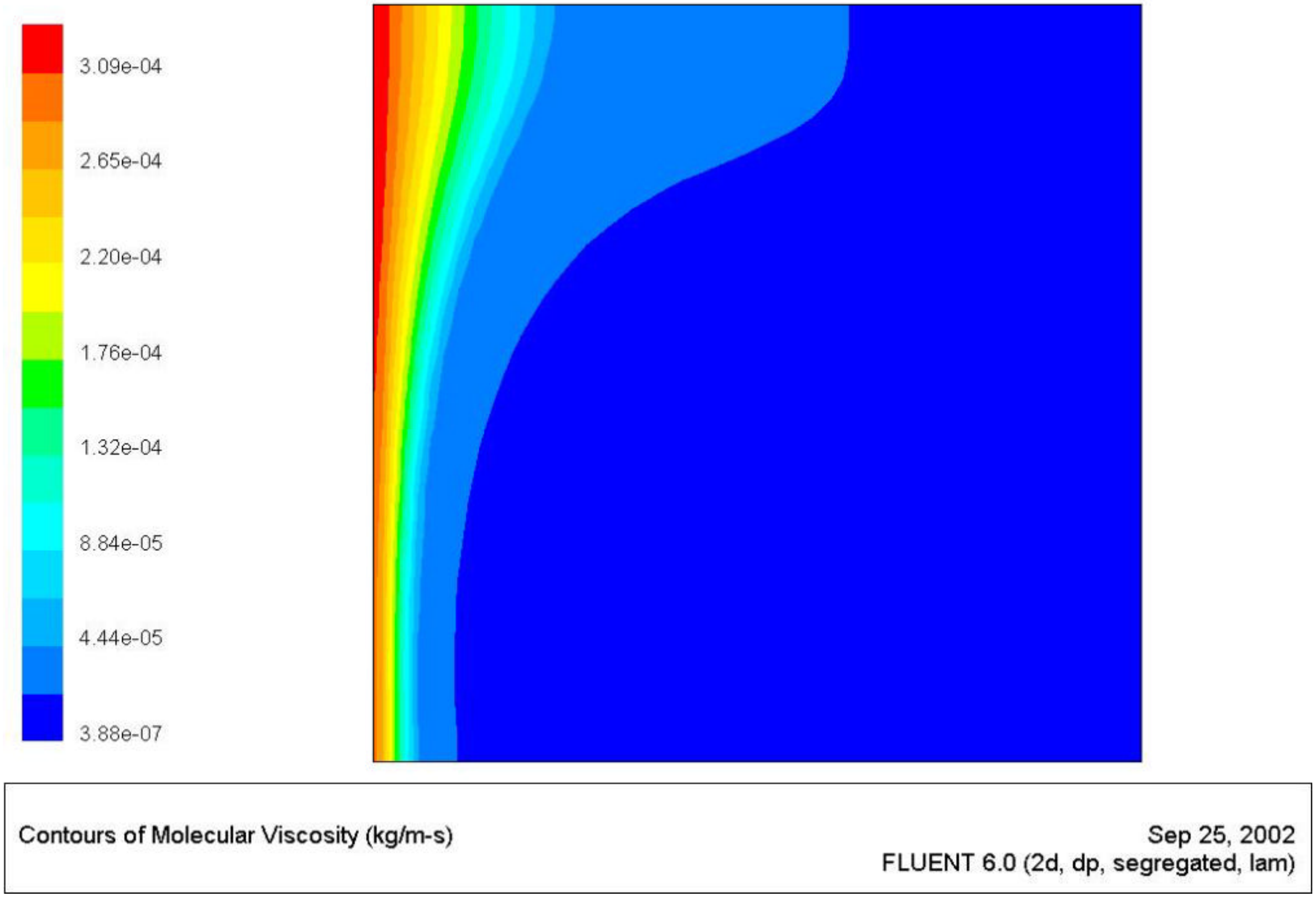
Buoyancy-driven flow in a square enclosure with variable viscosity



Buoyancy-driven flow in a square enclosure with variable viscosity



Buoyancy-driven flow in a square enclosure with variable viscosity



Summary

- Laminar flow problems with entrance region, seperation&reattachment and Bouyancy effects on FLUENT.
- Grid refinement performed for each problem.
- Good agreement with the previous computational and experimental studies.
- Recommendations for the solved benchmark problems with a clear problem statement, accurate predictions, and reliable experimental data (if available).

Future work

- A similar study for turbulent validation on FLUENT.
  - Backward-facing step problem with Re_H=13000, 20000.
  - Turbulent flow around a cylinder.
  - Falkner-Skan wedge (BL-stagnation) flows.
- 2D and 3D laminar and turbulent impinging jets (flat plate geometry) with heat transfer.